

Notes: Lin (RFS, 2020)

Bank Deposits and the Stock Market

Contents

1	Big picture	2
2	Data and measurement	3
2.1	Aggregate time-series data	3
2.2	County and ZIP-code deposit data	3
2.3	Stock market participation proxy	3
2.4	Bank-level data	3
2.5	Real activity data	4
3	Models and empirical specifications	5
3.1	Aggregate time-series specification	5
3.2	Simple county sensitivity regression	5
3.3	County-level panel regression	5
3.4	Within-bank specification	5
3.5	Within-county ZIP-code specification	5
3.6	Bank-level funding and lending specification	6
3.7	Real-effects specification	6
4	Identification and empirical design	7
4.1	Identification problem	7
4.2	Why geographic stock participation helps	7
4.3	Why within-bank tests matter	7
4.4	Why within-county ZIP-code tests matter	7
4.5	Why the IV test matters	7
4.6	Limits	8
5	Tables and figures	9
5.1	Figure 1 and Table 1: aggregate relation and summary statistics	9
5.2	Figures 2–3: measuring local stock market participation	10
5.3	Figures 4–5: household portfolio substitution	11
5.4	Table 2 and Figure 6: aggregate deposit growth and county participation deciles	12
5.5	Tables 3–4: county, within-bank, and within-county evidence	13
5.6	Tables 5–6: IV and robustness checks	14
5.7	Tables 7–8: bank lending and real effects	15
5.8	Appendix visuals: large banks, validation, and employment robustness	16
6	Short summary	18
7	Conclusion	18
7.1	Core conclusion	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** Can stock market booms reduce bank funding by pulling household money out of deposits and into stocks, and can that affect bank lending and real activity?
- **Core channel.** Stock market performance shifts household portfolio demand. When returns are high, households in high stock-participation areas reduce demand for deposits. This drains retail deposit funding from banks.
- **Bank balance-sheet mechanism.** If deposits are a stable and low-cost source of funding, reduced deposit growth forces banks to contract lending or to rely on more expensive alternatives.
- **Real effect.** Counties served by banks whose depositor bases are more stock-active experience weaker small business lending and weaker employment growth in externally finance-dependent sectors after stock market booms.
- **Identification idea.** The paper uses geographic heterogeneity in household stock market participation. The key measure is the local dividend-income ratio from IRS tax data.
- **Main county result.** Deposit growth is more negatively related to stock returns in counties with higher dividend-income ratios. The preferred weighted estimate implies that a one-standard-deviation annual stock return of about 20 percentage points reduces aggregate deposit growth by roughly 1 percentage point.
- **Within-bank evidence.** Across branches of the same bank, deposit growth falls more in counties with higher stock participation during stock market booms. This helps separate deposit demand from bank-wide credit demand or funding policy.
- **Within-county evidence.** Across ZIP codes inside the same county, high stock-participation ZIP codes have slower deposit growth during stock booms, even with county-by-year fixed effects.
- **Bank-level result.** Banks financed by more stock-active depositor bases see slower deposit growth when stock returns are high. Loan growth falls with a lag, especially C&I and real estate loans.
- **Economic takeaway.** Equity-market booms can crowd out bank deposits. This creates a transmission channel from capital markets to bank credit supply and local real activity.

2 Data and measurement

2.1 Aggregate time-series data

- The aggregate time-series analysis uses quarterly data from 1984 to 2014.
- Domestic office deposit growth and loan growth come from the FDIC Quarterly Banking Profile.
- Stock returns are cumulative returns on the CRSP value-weighted index.
- Controls include GDP growth, changes in the federal funds rate, and house price growth.
- Investor sentiment is from the Baker-Wurgler index, and stock-market inflow is measured as net flows into equity mutual funds scaled by lagged aggregate stock-market capitalization.

Interpretation. Aggregate time-series evidence motivates the channel: deposit growth is lower when the stock market performs well. But aggregate correlation alone cannot identify whether the mechanism is deposit demand, loan demand, or bank supply.

2.2 County and ZIP-code deposit data

- County-level deposits are constructed by aggregating branch deposits from the FDIC Summary of Deposits.
- The county sample covers 1994 to 2015.
- ZIP-code-level deposits are constructed by matching branch ZIP codes from Summary of Deposits to IRS ZIP-code tax data.
- Deposit growth is usually measured from June 30 of year $t - 1$ to June 30 of year t .

Interpretation. County and ZIP-code data allow the paper to compare places with different household stock participation but common macro shocks.

2.3 Stock market participation proxy

- The central proxy is

$$Div_ratio_{i,t} = \frac{\text{taxable dividend income}_{i,t}}{\text{total adjusted taxable income}_{i,t}}.$$

- IRS Statistics of Income provide dividend income and taxable income at county and ZIP-code levels.
- The measure varies substantially across regions and is designed to proxy for local household stock ownership.
- The paper validates this proxy by comparing state-level IRS measures with stock participation in Hong, Kubik, and Stein (2004).

Interpretation. The identification is strongest if *Div_ratio* captures stock-market exposure of local households rather than unrelated local fundamentals. The paper therefore includes controls, fixed effects, within-bank comparisons, within-county comparisons, and IV tests.

2.4 Bank-level data

- Bank balance-sheet variables come from quarterly Call Reports from 1994 to 2015.
- The main bank-level stock-participation measure is a deposit-weighted average of county dividend-income ratios across counties where the bank takes deposits.
- Bank outcomes include total deposits, checking deposits, time and savings deposits, wholesale funding, total loans, C&I loans, real estate loans, and deposit rates.
- Deposit rates come from Ratewatch for CDs and money market accounts.

Interpretation. The bank-level data test whether the local deposit-demand shock becomes a bank funding shock and then a lending shock.

2.5 Real activity data

- Small business lending comes from Community Reinvestment Act data.
- Employment data come from County Business Patterns and industry-level employment data.
- External finance dependence is measured at the industry level using a Rajan-Zingales style approach based on Compustat data.
- The real-effects tests compare counties based on the stock-market participation of the banks that fund and lend in those counties.

Interpretation. The real-effects section asks whether the funding shock matters outside bank balance sheets. The strongest prediction is for bank-dependent borrowers and externally finance-dependent employment sectors.

3 Models and empirical specifications

3.1 Aggregate time-series specification

The aggregate quarterly regression is

$$\Delta Depo_t = \alpha + \beta_1 ret_t + \beta_2 ret_{t-1} + \gamma_1 X_t + \gamma_2 X_{t-1} + \varepsilon_t. \quad (1)$$

- $\Delta Depo_t$ is aggregate domestic deposit growth from quarter $t - 1$ to t .
- ret_t and ret_{t-1} are current and lagged quarterly stock returns.
- X_t includes GDP growth, federal funds rate changes, and house price growth.

Interpretation. A negative β means deposit growth slows when the stock market is strong. This establishes the aggregate pattern but does not by itself identify the household portfolio channel.

3.2 Simple county sensitivity regression

For each county, the paper first estimates

$$\Delta Depo_{i,t} = \alpha_i + \beta_i ret_t + \varepsilon_{i,t}. \quad (2)$$

Counties are then sorted into dividend-income-ratio deciles and the average β_i is plotted by decile.

Interpretation. If the household portfolio channel is correct, high stock-participation counties should have more negative β_i coefficients. Figure 6 shows this nearly monotonic pattern.

3.3 County-level panel regression

The main county-level specification is

$$\Delta Depo_{i,t} = \alpha_i + \beta_1 Div_ratio_{i,t-1} \times ret_t + \beta_2 Div_ratio_{i,t-1} + \gamma X_{i,t-1} + \mu_t + \varepsilon_{i,t}. \quad (3)$$

- μ_t are year fixed effects, so the level of stock returns is absorbed by time effects.
- The coefficient of interest is β_1 .
- Controls include lagged income, population, income growth, and population growth.

Interpretation. The regression uses cross-county differences in stock participation to identify how a common stock-market shock affects deposit growth differently across places.

3.4 Within-bank specification

For multicounty banks, the paper estimates

$$\Delta Depo_{i,b,t} = \alpha_i + \beta_1 Div_ratio_{i,t-1} \times ret_t + \beta_2 Div_ratio_{i,t-1} + \gamma X_{i,t-1} + u_{b,t} + \varepsilon_{i,b,t}, \quad (4)$$

where $u_{b,t}$ are bank-by-year fixed effects.

Interpretation. Bank-by-year fixed effects compare deposits across counties within the same bank in the same year. This removes bank-wide funding, lending, and balance-sheet shocks.

3.5 Within-county ZIP-code specification

The ZIP-code specification is

$$\Delta Depo_{i,j,t} = \alpha + \beta_1 Div_ratio_{i,j,t-1} \times ret_t + \beta_2 Div_ratio_{i,j,t-1} + \beta_3 Div_ratio_{i,j,t-1} \times sentiment_t + u_{i,t} + \varepsilon_{i,j,t}. \quad (5)$$

- j indexes ZIP code and i indexes county.
- $u_{i,t}$ are county-by-year fixed effects.

Interpretation. County-by-year fixed effects compare ZIP codes inside the same county in the same year. This is a strong test against county-wide loan demand or local economic shocks.

3.6 Bank-level funding and lending specification

The bank-level regression is

$$\Delta Y_{i,t} = \alpha_i + \beta_1 Div_ratio_{i,t-1} \times ret_t + \beta_2 Div_ratio_{i,t-1} \times ret_{t-1} + \beta_3 X_{i,t-1} + \mu_t + \varepsilon_{i,t}. \quad (6)$$

- Y is deposit growth, loan growth, wholesale funding growth, or deposit rate changes.
- Div_ratio is the deposit-weighted stock-participation measure of a bank's deposit base.
- X includes bank controls such as assets, equity ratio, and income ratio.

Interpretation. The model tests whether household deposit demand shocks translate into bank funding and credit supply decisions.

3.7 Real-effects specification

The county real-effects regression is

$$\Delta Emp_{i,t} = \alpha_i + \beta_1 BankDiv_ratio_{i,t-1} \times ret_t + \beta_2 BankDiv_ratio_{i,t-1} \quad (7)$$

$$+ \beta_3 Div_ratio_{i,t-1} \times ret_t + \beta_4 Div_ratio_{i,t-1} + \gamma_1 X_{i,t-1} + \gamma_2 X_{i,t-1} \times Year2008 + \mu_t + \varepsilon_{i,t}. \quad (8)$$

Interpretation. The key variable is not the county's own household stock participation but the participation of the deposit bases of banks serving that county. This isolates credit-supply exposure from local household wealth effects.

4 Identification and empirical design

4.1 Identification problem

The aggregate relation between stock returns and deposit growth can be explained by multiple mechanisms:

- households shift money from deposits to stocks;
- public firms issue equity during booms and reduce loan demand;
- banks adjust deposit supply because credit demand changes;
- local economic fundamentals move with stock returns.

Interpretation. The empirical design is built to separate household deposit demand from bank loan demand and macroeconomic fundamentals.

4.2 Why geographic stock participation helps

- Stock returns are common across counties.
- Household stock participation varies across counties and ZIP codes.
- If the household portfolio channel is active, high-participation areas should react more strongly to the same stock-market return.
- Loan-demand explanations do not naturally predict this exact interaction with local household dividend income.

Interpretation. The interaction design turns a common aggregate shock into a heterogeneous local exposure test.

4.3 Why within-bank tests matter

- A bank can operate in many counties.
- Bank-wide shocks, lending policies, and funding conditions are absorbed by bank-by-year fixed effects.
- The remaining comparison is among branches of the same bank operating in places with different household stock participation.

Interpretation. If the result were driven by bank-wide loan demand or supply, it should disappear inside the same bank-year. It does not.

4.4 Why within-county ZIP-code tests matter

- County-by-year fixed effects absorb any county-wide shock in a given year.
- The estimation compares nearby ZIP codes within the same county.
- This helps rule out county-level macro shocks and broad local loan-demand shocks.

Interpretation. The ZIP-code test is noisy because depositors may bank outside their ZIP code, but it is valuable because it uses very local within-county variation.

4.5 Why the IV test matters

- The IV uses the fraction of adults with a bachelor's degree in 1990 as an instrument for the dividend-income ratio.
- Education is measured before the main sample and is correlated with stock participation.
- The IV results produce larger negative coefficients on $Div_ratio \times ret$.

Interpretation. The IV does not prove perfect exogeneity, but it addresses the concern that current dividend income is mechanically related to omitted current deposit-growth determinants.

4.6 Limits

- Dividend income is a proxy for stock participation, not a direct measure of household portfolios.
- Stock returns are not randomly assigned and may reflect fundamentals or sentiment.
- ZIP-code data are noisy because people may hold deposits outside their residential ZIP code.
- The real-effects tests use aggregate county outcomes, so the lending-to-employment channel is not observed firm by firm.

Interpretation. The paper's strength is the accumulation of consistent evidence across aggregate, county, ZIP, branch, bank, and real-activity samples.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* structure. Adjacent visuals are grouped to save space and make comparisons faster. All visuals are directly cropped from the original paper.

5.1 Figure 1 and Table 1: aggregate relation and summary statistics

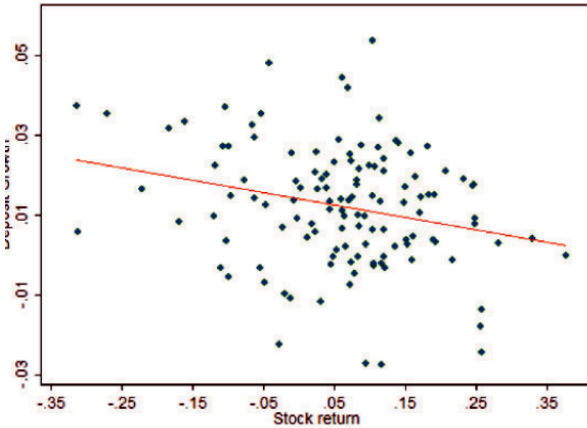
Read: Figure 1:

- The scatter plot shows a negative relation between quarterly deposit growth and stock returns from 1984 to 2014.
- Deposit growth tends to be weaker or negative when stock returns are high.
- This is surprising if both deposits and stock returns were simply procyclical.

Read: Table 1:

- Aggregate quarterly deposit growth averages 1.2%, while quarterly stock returns average 3.0%.
- County-level annual deposit growth averages 3.6%; the average county dividend-income ratio is 1.6%.
- The bank-level panel is large, with more than 648,000 bank-quarter observations for many outcomes.

Economic message. The aggregate negative relation motivates the paper. The summary statistics show that the analysis uses multiple layers of data: aggregate time series, counties, ZIP codes, banks, and real-activity outcomes.



(a) Figure 1: deposit growth and stock returns.

Table 1
Summary statistics

A. Time-series data (quarterly)

	Mean	Median	SD	Min	Max	No. of obs.
<i>Deposit growth</i>	0.012	0.013	0.015	-0.028	0.054	123
<i>Loan growth</i>	0.011	0.014	0.015	-0.028	0.043	123
<i>Stock return</i>	0.030	0.039	0.085	-0.237	0.213	123
<i>FF rate</i>	0.042	0.047	0.030	0.001	0.113	123
<i>GDP growth</i>	0.007	0.007	0.006	-0.021	0.019	123
<i>HP growth</i>	0.009	0.010	0.012	-0.031	0.038	123
<i>Investor sentiment</i>	0.296	0.199	0.616	-0.820	2.760	123
<i>Flow into stocks</i>	0.002	0.002	0.003	-0.007	0.010	123

B. County and ZIP code data (annual)

County-year data

<i>Div_ratio</i>	0.016	0.014	0.009	0.003	0.063	65,275
<i>Deposit growth</i>	0.036	0.031	0.071	-0.257	0.447	65,275
<i>Ipc</i>	27,158	25,582	9,503	5,082	189,550	65,275
<i>Pop</i>	94,600	25,844	300,927	421	10,053,995	65,275
<i>Ipc growth</i>	0.038	0.038	0.049	-0.159	0.256	65,275
<i>Pop growth</i>	0.005	0.004	0.015	-0.038	0.064	65,275
<i>HHI</i>	0.319	0.255	0.209	0.000	1.000	65,275
<i>Emp growth</i>	0.003	0.006	0.058	-0.272	0.281	43,734
<i>Emp-dependent growth</i>	0.000	0.004	0.115	-0.659	1.746	43,734
<i>Emp-nondependent growth</i>	0.006	0.007	0.090	-0.642	0.503	43,734
<i>Small loan growth</i>	0.014	0.017	0.348	-1.018	1.144	43,734
<i>HP growth</i>	0.024	0.023	0.056	-0.445	0.553	43,734

County data in 1990

<i>Age</i>	34.4	34.3	3.6	20.0	55.4	3,117
<i>Bachelor's degree</i>	0.134	0.118	0.641	0.037	0.534	3,117

ZIP-code-year data

<i>Div_ratio</i>	.0144	.0100	.0149	.0002	.1045	435,094
<i>Deposit growth</i>	0.068	0.032	0.273	-0.638	2.591	435,094

(Continued)

(b) Table 1, Panels A-B: aggregate, county, and ZIP summary statistics.

Table 1
Continued

C. Bank-level data (quarterly)

<i>Div_ratio</i>	0.017	0.016	0.008	0.002	0.063	648,5
<i>Stock return</i>	0.028	0.039	0.088	−0.237	0.213	648,5
<i>Deposit growth</i>	0.019	0.012	0.061	−0.129	0.403	648,5
<i>Checking growth</i>	0.022	0.018	0.174	−0.684	0.782	648,5
<i>Time&savings growth</i>	0.019	0.011	0.064	−0.135	0.412	648,5
<i>Wholesale growth</i>	0.023	0.000	0.522	−2.053	2.213	429,4
<i>Loan growth</i>	0.021	0.016	0.062	−0.177	0.452	648,5
<i>C&I Loan growth</i>	0.020	0.013	0.149	−0.511	0.700	648,5
<i>RE Loan growth</i>	0.025	0.016	0.070	−0.155	0.473	648,5
<i>ln(assets)</i>	11.707	11.564	1.331	7.313	21.463	648,5
<i>Deposit ratio</i>	0.838	0.856	0.077	0.083	0.930	648,5
<i>Equity ratio</i>	0.106	0.097	0.037	0.054	0.666	648,5
<i>Income ratio</i>	0.006	0.005	0.006	−0.028	0.028	648,5
<i>Rate_12mcd10k</i>	2.350	1.980	1.762	0.001	12.588	331,5
<i>Rate_mm25k</i>	1.294	0.950	1.185	0.001	9.356	317,7

Panel A reports the summary statistics of quarterly aggregate data from 1984 to 2014. Deposit and loan growth

Figure 2: Table 1, Panel C: bank-level summary statistics.

5.2 Figures 2–3: measuring local stock market participation

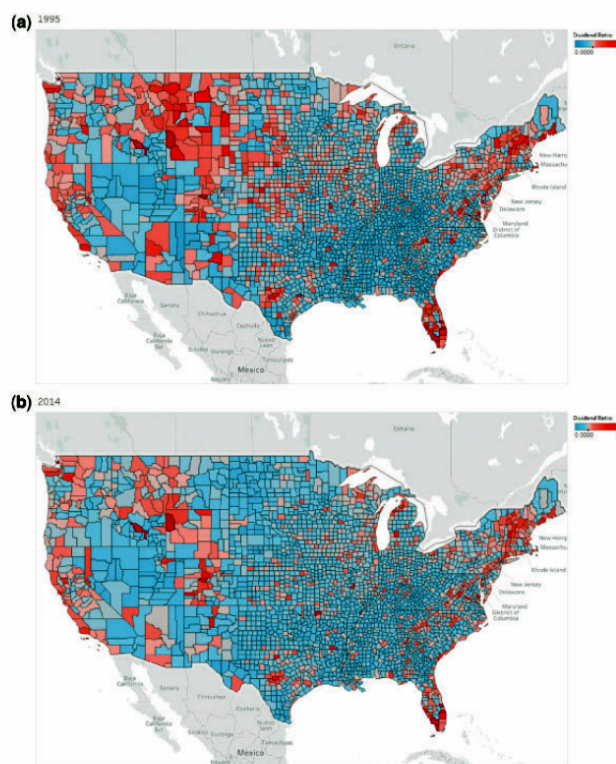
Read: Figure 2:

- The maps show county-level dividend-income ratios in 1995 and 2014.
- There is large geographic heterogeneity in household stock participation.
- The distribution changes over time, but persistent participation differences remain.

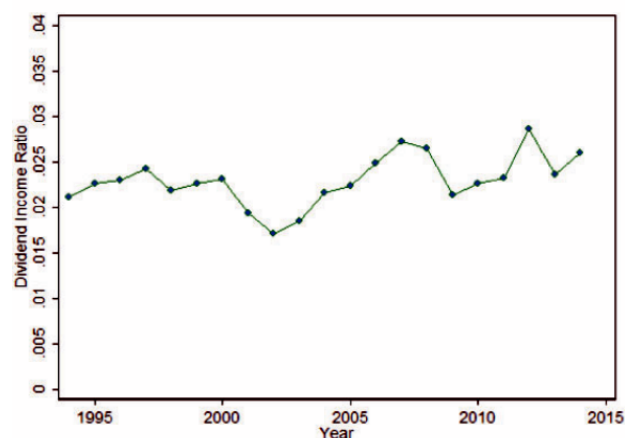
Read: Figure 3:

- The national dividend-income ratio fluctuates over time but remains in a narrow range around roughly 2%.
- This confirms that the proxy is meaningful over the paper’s main sample.

Economic message. The paper’s core identification comes from comparing areas with high and low stock-market participation when the same aggregate stock return occurs.



(a) Figure 2: dividend-income ratio maps.



(b) Figure 3: aggregate dividend-income ratio.

5.3 Figures 4–5: household portfolio substitution

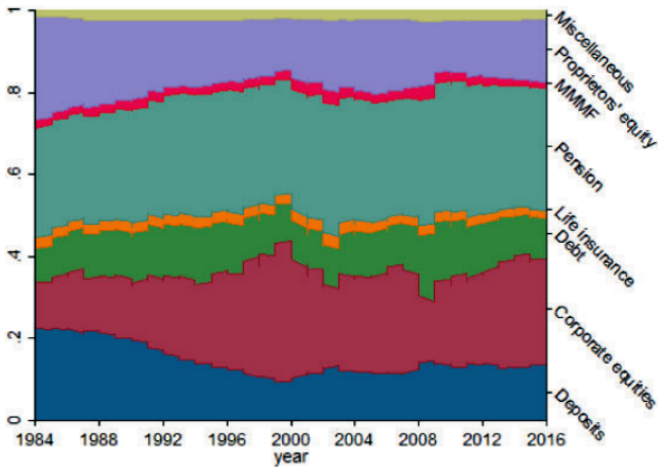
Read: Figure 4:

- Household financial assets are split across deposits, corporate equities, pensions, insurance, debt/MMMF, proprietors' equity, and miscellaneous assets.
- Deposit shares and corporate equity shares tend to move in opposite directions.
- The 1990s equity boom coincides with a shrinking deposit share; the early 2000s and crisis periods show the opposite.

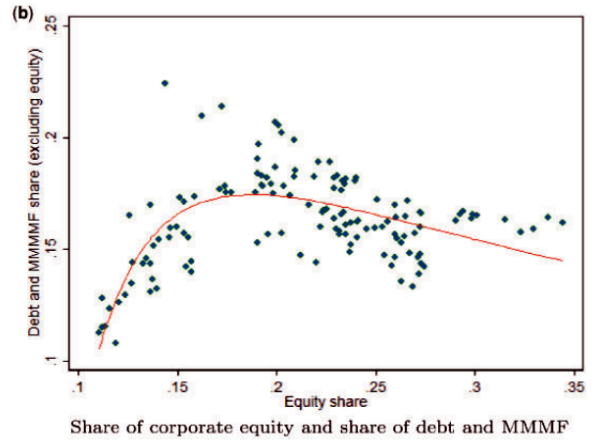
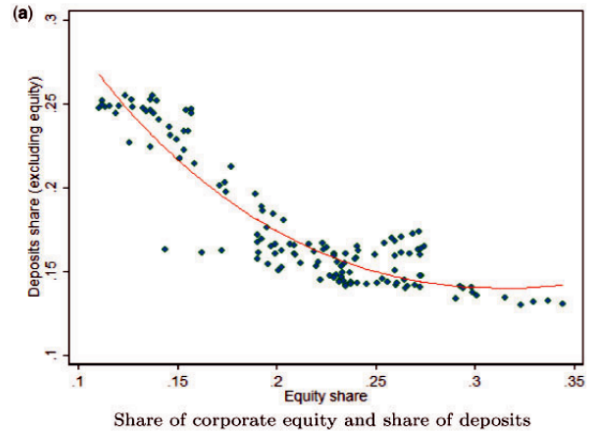
Read: Figure 5:

- Panel A shows a tight negative relation between the corporate-equity share and the deposit share excluding equity.
- The paper reports a correlation of about -0.87 for this relation.
- Panel B shows a much weaker and nonlinear relation between equity share and debt/MMMF share.

Economic message. The household balance-sheet evidence supports substitution between stocks and bank deposits, not simply between stocks and all safe or fixed-income assets.



(a) Figure 4: household financial asset shares.



(b) Figure 5: equity share versus deposits and debt/MMMF.

5.4 Table 2 and Figure 6: aggregate deposit growth and county participation deciles

Read: Table 2:

- Current and lagged stock returns are negatively related to aggregate deposit growth.
- With controls, the coefficients are -0.030 for current return and -0.037 for lagged return.
- Stock returns are positively related to flow into equity mutual funds.
- Loan growth also falls with lagged stock returns, although this aggregate evidence is not yet causal.

Read: Figure 6:

- Counties are sorted into deciles by dividend-income ratio.
- The deposit-growth sensitivity to stock returns becomes more negative almost monotonically across deciles.
- High stock-participation counties experience the strongest deposit outflows during stock booms.

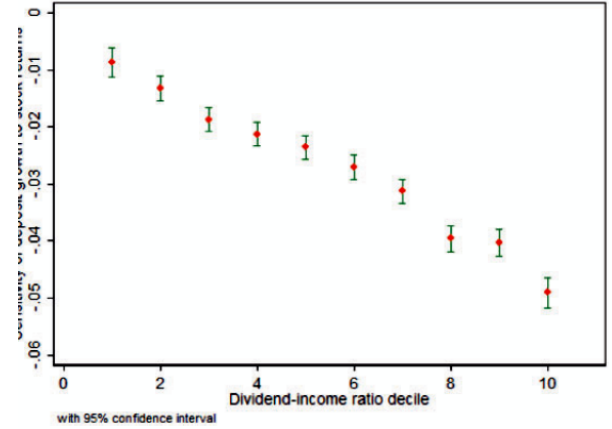
Economic message. Table 2 documents the aggregate pattern. Figure 6 gives the first cross-sectional evidence that the pattern is linked to household stock participation.

Table 2
Deposit growth and the stock market

	Deposit growth (1)	Deposit growth (2)	Loan growth (3)	Flow into stock market (4)
$Return_t$	-0.023 (0.016)	-0.030* (0.016)	-0.018 (0.013)	0.013*** (0.003)
$Return_{t-1}$	-0.038** (0.016)	-0.037** (0.016)	-0.030** (0.013)	0.009*** (0.003)
$GDP\ growth_t$		-0.042 (0.270)	0.559** (0.214)	0.079 (0.050)
$GDP\ growth_{t-1}$		0.113 (0.257)	0.403* (0.204)	0.038 (0.048)
$\Delta FRate_t$		-0.489* (0.254)	0.180 (0.201)	0.013 (0.047)
$\Delta FRate_{t-1}$		0.370 (0.252)	0.461** (0.200)	-0.033 (0.047)
$HP\ growth_t$		-0.011 (0.168)	0.256* (0.133)	-0.013 (0.031)
$HP\ growth_{t-1}$		0.233 (0.172)	0.175 (0.136)	0.053 (0.032)
R-squared	.064	.132	.416	.328
N	123	123	123	123

The dependent variable is quarterly aggregate deposit growth in Columns 1 and 2, and loan growth in Column 3 from 1984 to 2014. In the last column, the dependent variable is the net dollar inflows in each quarter into equity mutual funds, scaled by lagged aggregate capitalization of the U.S. stock market. $Return$ is the quarterly return of the value-weighted index from CRSP. $FRate$ is the effective Federal funds rate (not seasonally adjusted) from the Federal Reserve Bank of St. Louis. $GDP\ growth$ is the quarterly growth rate of the U.S. GDP from the Federal Reserve Bank of St. Louis. $HP\ growth$ is the quarterly growth rate of the U.S. residential house price index from the FHFA. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 2: aggregate deposit growth and stock returns.



(b) Figure 6: sensitivity by dividend-income decile.

5.5 Tables 3–4: county, within-bank, and within-county evidence

Read: Table 3:

- The county-level interaction $Div_ratio \times ret$ is negative in all specifications.
- Column 3, the population-weighted specification, estimates -2.32.
- The within-bank specifications remain negative: -1.47 and -1.58.
- Investor sentiment interacted with dividend income is also negative.

Read: Table 4:

- ZIP-code-level deposit growth also reacts more negatively to stock returns in high-dividend-income areas.
- With county-by-year fixed effects, the coefficient is smaller but still negative at -0.75.
- This helps rule out county-level shocks.

Economic message. The results survive stronger comparisons: within the same bank and within the same county. This is important because it weakens loan-demand and local-macro alternatives.

Table 3
Deposit growth and stock market participation at the county level

	County-level					Within-bank	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Div_ratio</i> × <i>ret</i>	-1.40** (0.52)	-1.36** (0.51)	-2.32** (1.02)	-1.61*** (0.51)		-1.47** (0.69)	-1.58** (0.50)
<i>Div_ratio</i> × <i>ret_above</i>					-1.85*** (0.61)		
<i>Div_ratio</i> × <i>ret_below</i>					-1.21* (0.69)		
<i>Div_ratio</i> × <i>sentiment</i>				-0.35*** (0.11)	-0.33*** (0.11)		-0.46** (0.18)
<i>Div_ratio</i>	0.17 (0.10)	0.18 (0.13)	0.44** (0.18)	0.27** (0.12)	0.30** (0.12)	-0.05 (0.14)	-0.28 (0.28)
<i>ln(Pop)</i>	-0.00 (0.00)	-0.03* (0.02)	0.00** (0.00)	-0.00 (0.00)	-0.00 (0.00)	0.02*** (0.00)	-0.04** (0.02)
<i>ln(Ipc)</i>	0.03*** (0.01)	0.04** (0.02)	0.03*** (0.01)	0.03*** (0.01)	0.03*** (0.01)	0.06*** (0.01)	0.06* (0.03)
<i>Popgrowth</i>	0.80*** (0.08)	0.51*** (0.08)	1.18*** (0.14)	0.80*** (0.08)	0.80*** (0.08)	1.08*** (0.12)	0.52** (0.14)
<i>Ipcgrowth</i>	0.15*** (0.03)	0.13*** (0.03)	0.18*** (0.05)	0.15*** (0.03)	0.15*** (0.03)	0.10** (0.04)	0.09** (0.04)
Year FE	Yes	Yes	Yes	Yes	Yes	No	No
County FE	No	Yes	No	No	No	No	Yes
Population weighted	No	No	Yes	No	No	No	No
Bank×year FE	No	No	No	No	No	Yes	Yes
State×year FE	No	No	No	No	No	Yes	Yes
R-squared	.066	.122	.119	.067	.067	.308	.318
N	65,275	65,275	65,275	65,275	65,275	195,604	195,566

In the first five columns, the dependent variable is deposit growth at the county level from June 30 of year $t-1$ to June 30 of year t . In the last two columns, the dependent variable is deposit growth at the bank-county level. *re* is the annual return of CRSP value-weighted index over the same period for which deposit growth is measured. *ret_above* is equal to *ret* if *ret* is above the sample median, and zero otherwise. *ret_below* is equal to *ret* if *re*

(a) Table 3: county-level and within-bank results.

Table 4
Deposit growth and stock market participation: Within-county analysis

	(1)	(2)
<i>Div_ratio</i> × <i>ret</i>	-1.37** (0.55)	-0.75* (0.38)
<i>Div_ratio</i> × <i>sentiment</i>	-0.25* (0.12)	-0.32*** (0.08)
Controls	Yes	No
Year FE	Yes	No
County×year FE	No	Yes
R-squared	.015	.125
N	435,094	425,695

The dependent variable is ZIP-code-level deposit growth. *Div_ratio* is the total dividend income over total income in a ZIP code. *ret* is the annual return of the CRSP value-weighted index over the same period for which deposit growth is measured. County controls in Column 1 include *Div_ratio*, lagged income per capita, lagged population, income per capita growth, and population growth. Table 1 defines all other variables. Standard errors are two-way clustered by year and county. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 4: within-county ZIP-code results.

5.6 Tables 5–6: IV and robustness checks

Read: Table 5:

- Panel A shows that education and age are positively related to the dividend-income ratio.
- Panel B instruments dividend-income ratio with the 1990 bachelor's-degree share.
- The IV coefficients on *Div_ratio* × *ret* are -4.07 and -4.76, larger than OLS estimates.

Read: Table 6:

- The main interaction remains negative after controlling for monetary policy channels, stock-return volatility, alternative stock-participation measures, and mutual fund inflows.
- The return-volatility interaction is not the main driver.
- The flow-into-stock-market specification also supports the channel.

Economic message. The robustness checks show that the result is not only an artifact of the dividend-income measure, monetary policy, stock volatility, or county fundamentals.

Table 5
Determinants of stock market participation and the IV estimation

A. Determinants of stock participation

	(1)	
$\ln(Pop)$	-0.029* (0.017)	
$\ln(Ipc)$	0.500*** (0.090)	
$Bachelor_{1990}$	0.064*** (0.004)	
Age_{1990}	0.089*** (0.005)	
HHI	0.109 (0.073)	
Year FE	Yes	
R-squared	.385	
N	65,275	
<i>B. IV estimation</i>		
	(1)	(2)
$Div_ratio \times ret$	-4.07** (1.86)	-4.76* (1.29)
Div_ratio	1.89*** (0.32)	2.11* (0.27)
Controls	Yes	Yes
Year FE	Yes	Yes
Income and industry interactions	No	Yes
R-squared	.045	.048
N	65,275	65,275

In panel A, the dependent variable is the ratio of taxable dividend income over total adjusted taxable income measured in percentage points. Panel B reports the results for the IV estimation, where the dependent variable is deposit growth. The instrument for dividend income ratio is the fraction of population of age 25 or older.

(a) Table 5: determinants and IV estimation.

Table 6
Alternative channels and robustness checks

	(1)	(2)	(3)	(4)	(5)
$Div_ratio \times ret$	-1.32*** (0.40)	-1.53*** (0.47)	-1.53*** (0.50)	-0.18*** (0.05)	
$Div_ratio \times vol$			-2.84 (3.91)		
$Div_ratio \times inflow$					-20.37** (5.33)
$Div_ratio \times \Delta fed_rate$		5.17 (9.13)			
$HHI \times \Delta fed_rate$		-1.16*** (0.27)			
HHI		0.03*** (0.00)			
$\ln(Pop) \times ret$	-0.00 (0.03)	-0.00 (0.03)	0.03 (0.05)	0.02 (0.03)	-0.01 (0.03)
$\ln(Ipc) \times ret$	-0.01 (0.01)	-0.01* (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
R-squared	.067	.074	.067	.067	.067
N	65,275	65,275	65,275	65,196	65,275

The dependent variable is county deposit growth in all five columns. ret is the annual return of the CRSP value weighted index. Δfed_rate is the change in the effective Federal funds rates over the same period. vol is the standard deviation of monthly CRSP returns over the same period. Div_ratio is the ratio of taxable dividend income to total adjusted taxable income from the IRS, except for in Column 4, where div_ratio is the fraction of income tax returns that report dividend income in a county. The number of dividend tax returns in 2004 where it becomes available is used for the pre-2004 years. In Column 5, $inflow$ is defined as aggregate net inflow into equity mutual funds during the same period, scaled by the lagged aggregate capitalization of the U.S. stock market. HHI is the Herfindahl index of deposit market share in a given county in a given year. Controls include div_ratio , lagged income per capita, lagged population, income per capita growth, population growth, and

(b) Table 6: alternative channels and robustness.

5.7 Tables 7–8: bank lending and real effects

Read: Table 7:

- Banks exposed to high-participation depositor bases experience slower total deposit growth when contemporaneous stock returns are high.
- Time and savings deposits show the strongest deposit response.
- Loan growth responds with a lag: total loans, C&I loans, and real estate loans fall following stock booms in high-participation areas.
- Money-market deposit rates rise, consistent with banks trying to retain deposit funding.

Read: Table 8:

- Counties borrowing from banks exposed to high-participation deposit bases experience weaker small business lending.
- Total employment effects are small and statistically weak.
- Employment in externally finance-dependent sectors falls significantly; nondependent sectors do not show the same effect.

Economic message. The deposit-demand shock becomes a bank credit-supply shock and then a real shock to bank-dependent firms. The sector split is central: the effect appears where bank finance should matter most.

Table 7
Deposit and loan growth at the bank level

A. No bank fixed effects

	Deposits			Wholesale		Loans			Rates	
	(1) Total	(2) Checking	(3) Time&savings	(4) Wholesale	(5) Total	(6) C&I	(7) RE	(8) 12M CD	(9) 25k MM	
$Div_ratio \times ret_t$	-1.27*** (0.45)	-0.62 (0.95)	-1.26*** (0.43)	-0.35 (2.15)	-0.21 (0.36)	-0.19 (0.34)	-0.42 (0.34)	1.18 (1.92)	3.03* (1.54)	
$Div_ratio \times ret_{t-1}$	-0.32 (0.40)	1.35* (0.79)	-0.59 (0.39)	-0.47 (2.54)	-0.79* (0.46)	-1.05** (0.40)	-0.82* (0.44)	3.42 (2.97)	3.70** (1.20)	
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Bank FE	No	No	No	No	No	No	No	No	No	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
R-squared	.040	.052	.037	.032	.059	.012	.038	.665	.306	
N	648,502	648,502	648,502	429,426	648,502	648,502	648,502	325,024	311,207	
B. Bank fixed effects										
$Div_ratio \times ret_t$	-1.08*** (0.37)	-0.45 (0.89)	-1.06*** (0.34)	0.30 (2.14)	0.05 (0.32)	0.03 (0.34)	-0.14 (0.28)	1.26 (1.91)	3.06** (1.52)	
$Div_ratio \times ret_{t-1}$	-0.24 (0.35)	1.40* (0.76)	-0.50 (0.33)	-0.08 (2.47)	-0.64* (0.38)	-0.96** (0.38)	-0.65* (0.35)	3.39 (2.89)	3.63** (1.17)	
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
R-squared	.148	.071	.136	.053	.174	.047	.138	.666	.310	
N	648,369	648,369	648,369	429,060	648,369	648,369	648,369	325,010	311,189	

The dependent variable in Columns 1 to 7 is quarterly bank deposit growth or loan growth, and Columns 8 and 9 are the change in quarterly rates of the 12-month CDs of \$10,000 and the change in rates of \$25,000 money market accounts. The quarterly rates are computed as the average weekly deposit rates across branches within a bank. Div_ratio is the deposit-weighted dividend-income ratio in counties where a bank has deposits. ret_t and ret_{t-1} are the cumulative return of the CRSP value-weighted index in quarter t and quarter $t-1$, respectively. Control include Div_ratio , log bank assets, bank equity ratio, and bank income scaled by assets. Standard errors are

(a) Table 7: bank-level deposits, lending, and rates.

Table 8
Real effects of the deposit channel

	Small business lending		Emp: Total		Emp: Dependent		Emp: Nondependent	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Bank_Div_ratio \times ret$	-11.89*** (3.57)	-12.64*** (3.31)	-0.19 (0.40)	-0.34 (0.42)	-1.33* (0.69)	-1.68** (0.70)	-0.14 (0.68)	-0.13 (0.72)
$Bank_Div_ratio$	0.45 (0.83)	0.06 (2.08)	0.37*** (0.13)	0.15 (0.24)	0.63*** (0.18)	0.97** (0.40)	0.21 (0.16)	-0.34 (0.30)
$Own_Div_ratio \times ret$	3.03** (1.25)	3.31** (1.26)	0.16 (0.20)	0.21 (0.21)	-0.02 (0.28)	0.11 (0.35)	0.75* (0.40)	0.74* (0.40)
Own_Div_ratio	-0.67*** (0.11)	-0.79 (0.50)	-0.15* (0.08)	0.02 (0.09)	-0.17 (0.11)	0.05 (0.23)	-0.24*** (0.08)	-0.16 (0.19)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	No	Yes	No	Yes	No	Yes	No	Yes
R-squared	.195	.211	.112	.173	.056	.095	.035	.066
N	43,711	43,711	43,711	43,711	43,711	43,711	43,711	43,711

In the first two columns, the dependent variable is the growth of small business loans from the end of calendar year $t-1$ to the end of calendar year t . In the next six columns, the dependent variable is the growth of count employment from March of calendar year t to March of calendar year $t+1$, for all sectors (Columns 3 and 4 sectors dependent on external finance (Columns 5 and 6), and sectors not dependent on external finance (Column 7 and 8). ret is the 5-quarter cumulative return of the CRSP value-weighted index from the last quarter of calendar year $t-1$ to the end of calendar year t . $Bank_Div_ratio$ is defined as the weighted average of bank dividend income ratios across all banks in a given county, using their lagged deposits shares as weights. Own_Div_ratio is the county's own dividend income ratio. The regressions control for deposits-weighted bank assets, equity ratio income ratio, deposit ratio, real estate loan ratio, the county's house price growth, and their interactions with the year dummy of 2008. Standard errors are two-way clustered by year and county. * $p < .1$; ** $p < .05$; *** $p < .01$

that the effect is small and not statistically significant for nondependent sectors. Taken together, the findings in this section support the hypothesis that stock

(b) Table 8: real effects of the deposit channel.

5.8 Appendix visuals: large banks, validation, and employment robustness

Read:

- Table A.I shows that the bank-level lending and funding results are similar among relatively large banks.
- Figure A.I validates the IRS dividend-income proxy using state-level stock participation measures from Hong, Kubik, and Stein.
- Table A.II confirms the employment result for externally finance-dependent industries in the sample with less employment imputation.

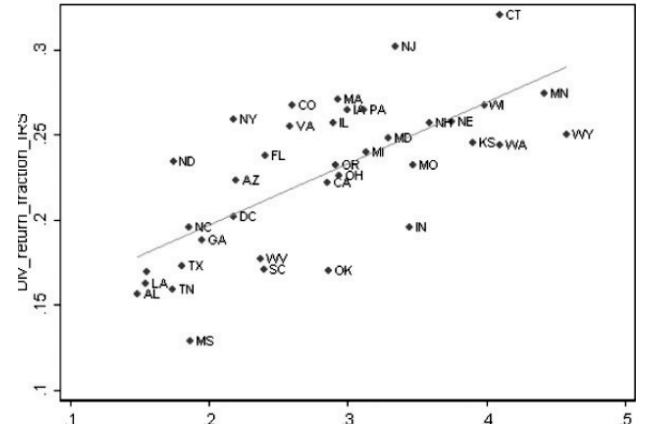
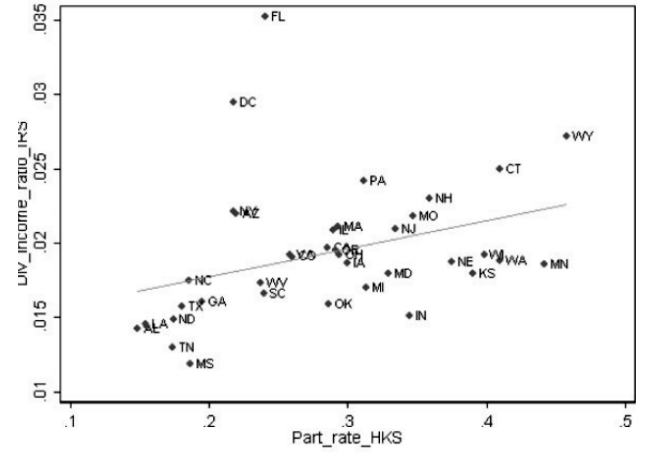
Economic message. The appendix reinforces the main identification strategy and shows that the real-effect result is not driven by small banks or by heavily imputed employment observations.

Table A.I
Deposit and loan growth of large banks

	Deposits			Wholesale	Loans			Rates	
	(1) Total	(2) Checking	(3) Saving&Time	(4) Wholesale	(5) Total	(6) C&I	(7) RE	(8) 12M CD	(9) 25k MM
$Div_ratio \times ret_t$	-0.88** (0.38)	-0.34 (1.25)	-0.78** (0.34)	-1.20 (2.14)	-0.26 (0.27)	-0.75* (0.45)	-0.37 (0.26)	0.20 (2.18)	2.19* (1.15)
$Div_ratio \times ret_{t-1}$	-0.21 (0.34)	1.18 (1.02)	-0.35 (0.35)	-0.69 (2.20)	-0.62* (0.35)	-1.07** (0.43)	-0.57 (0.35)	2.42 (3.18)	2.30** (1.10)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.039	.037	.037	.040	.063	.022	.054	.679	.315
N	215,929	215,929	215,929	186,097	215,929	215,929	215,929	118,122	115,625

The table reports the results of estimating Eq. (6) in the paper using the top third of sample banks with the largest average assets during the sample period. The dependent variable in Columns 1 to 7 is quarterly bank deposit growth or loan growth and in Columns 8 and 9 is the change in quarterly rates of the 12-month CDs of \$10,000 and the change in rates of \$25,000 money market accounts. The quarterly rates are computed as the average weekly deposit rates across branches within a bank. Div_ratio is the deposit-weighted dividend-income ratio in counties where a bank has deposits. ret_t and ret_{t-1} are the cumulative return of the CRSP value-weighted index in quarter t and quarter $t-1$, respectively. Controls include Div_ratio , log bank assets, bank equity ratio, and bank income scaled by lagged assets. Standard errors are two-way clustered by bank and time. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table A.I: large banks.



(b) Figure A.I: stock-participation proxy validation.

Table A.II
Employment of industries dependent on external finance

	(1)	(2)
$Bank\ Div_ratio \times ret$	-1.82** (0.80)	-3.91*** (0.78)
$Bank\ Div_ratio$	-0.51 (0.53)	-0.12 (0.56)
$Own\ Div_ratio \times ret$	0.91 (0.58)	1.38** (0.62)
$Own\ Div_ratio$	-0.12 (0.35)	-0.11 (0.37)
Controls	No	Yes
Year FE	Yes	Yes
County FE	Yes	Yes
R-squared	.201	.199
N	23,673	22,153

The dependent variable is the log growth of employment in external-finance-dependent industries from March of calendar year t to March of calendar year $t+1$. $Bank\ Div_ratio$ is defined as the weighted average of bank dividend-income ratios across all banks in a given county, using their lagged deposits shares as weights. ret is the 5-quarter cumulative return of the CRSP value-weighted index from the last quarter of calendar year $t-1$ to the end of calendar year t . $Own\ Div_ratio$ is the county's own dividend income ratio. The regressions control for deposits-weighted bank assets, equity ratio, income ratio, deposit ratio, real estate loan ratio, the county's house price growth, and their interactions with the year dummy of 2008. Standard errors are two-way clustered by year and county. * $p < .1$; ** $p < .05$; *** $p < .01$.

Figure 10: Table A.II: employment of externally finance-dependent industries.

6 Short summary

- The paper studies whether stock market booms reduce bank deposits through household portfolio rebalancing.
- Aggregate deposits grow more slowly when stock returns are high.
- Household balance-sheet data show that deposit shares and corporate-equity shares move in opposite directions.
- The key cross-sectional proxy for stock participation is the IRS dividend-income ratio.
- High dividend-income counties have more negative deposit-growth sensitivity to stock returns.
- The county-level interaction remains negative with county fixed effects, population weights, investor sentiment controls, and within-bank fixed effects.
- ZIP-code results with county-by-year fixed effects support the same mechanism at a more local level.
- IV tests using 1990 education as an instrument produce larger negative estimates.
- Robustness tests show that the result is not mainly driven by monetary policy, return volatility, or alternative local trends.
- At the bank level, exposure to high stock-participation deposit bases reduces deposit growth during stock booms.
- Loan growth falls with a lag, particularly C&I and real estate loans.
- Banks raise some deposit rates, consistent with attempts to retain deposit funding.
- Counties served by exposed banks have weaker small business lending.
- Employment falls in external-finance-dependent sectors but not in nondependent sectors.
- The main takeaway is that stock-market booms can crowd out bank-intermediated credit through household portfolio rebalancing.

7 Conclusion

7.1 Core conclusion

Lin shows that stock market booms can reduce households' demand for bank deposits. This matters because deposits are a key funding source for banks. When deposit growth slows, banks reduce lending, and the effect is transmitted to local employment in sectors that rely more on external finance.

7.2 Economic intuition

The mechanism is a funding-side crowding-out effect. Households in stock-active areas rebalance toward equities when the market performs well. Their reduced deposit demand lowers local bank deposit growth. Because deposits are relatively stable and cheap bank funding, banks cannot costlessly replace them with wholesale funding or equity. Lending therefore falls, especially after a lag.

7.3 Takeaway

The paper connects household portfolio choice, bank funding, credit supply, and real activity. It shows that stock market fluctuations do not only affect firms through equity issuance or wealth effects. They can also affect the banking system by changing where households place their liquid savings.